

30th May 2026

2:43 p.m.

I wrote the sieve of Eratosthenes code which will be used to prune candidate keys to ensure that they are indeed prime. Miller-Rabin test was used if the candidate wasn't divisible (~ 200 to 400 calls for Miller-Rabin tests). Then I followed it with implementing the extended Euclidean algorithm to make sure that all the functions in the RSA set-up work properly in terms for modular inverse calculations.

I also removed the old reports from the repository and moved the code which I used during the demonstration period to another file, now labelled, "old_cryptography_code_IGNORE.py". I put in the code that I worked on this week into the "cryptography_ui.py" file.

I worked on this week's stuff for ten hours.

Writing the functions in practice helped me to better understand the videos I'd watched during the past two weeks on those topics.

What I'm left with is the unit testing and test coverage which I plan on doing next week.

Could you also let me know if there's anything else I should add or if I'm missing something that you mentioned to me the last time we spoke about what other stuff I should add to my code (the call where you mentioned sieve of Eratosthenes, extended Euclidean algorithm etc)?